

MATH 3150, ANALYSIS I

GENERAL INFORMATION

- Instructor: Ovidiu Munteanu
- Email: ovidiu.munteanu@uconn.edu
- Lecture: TuTh 12:30PM - 1:45PM in MONT 225.
- Office Hours: Tu 10:00 am – 11:00 am and Th 11:00 am – 12:00 pm in MONT 434.
- Textbook: Kenneth A. Ross, Elementary Analysis: The Theory of Calculus (Undergraduate Texts in Mathematics), Springer, 2nd ed. 2013.

COURSE DESCRIPTION & GOALS

A rigorous introduction to the theory of real functions. The objective is to understand why theorems, such as the ones you learned in Calculus I, are true by thoroughly understanding their proofs. The topics we are going to cover include:

- (1) Basic properties of real numbers, including how existence of sup and inf distinguishes them from the naturals, integers and rationals.
- (2) Rigorous understanding of sequences: boundedness, monotonicity, convergence, limit, subsequences, the Bolzano-Weierstrass Theorem, and the Heine-Borel Theorem.
- (3) Formal definitions of limits and continuity of functions, including one-sided limits, infinite limits and limits at infinity.
- (4) Properties of continuous functions on closed bounded intervals: existence of maxima and minima, the Extreme Value Theorem, the Intermediate Value Theorem, uniform continuity.
- (5) Basic concepts of differential calculus: definition of derivative, Rolle's Theorem, the Mean Value Theorem and their consequences.
- (6) Basic concepts of integral calculus: Riemann sums, integrable functions, the Fundamental Theorem of Calculus.

GRADING

Homework. There will be 10 weekly assignments. Homework is always due on Thursday, except for Week 1, Week 6, and Week 11, when there is no homework. The first assignment is due on Thursday, September 10.

It is permitted to discuss homework with other students, but you must write it on your own. Late homework is not acceptable.

Homework will be posted and graded on HuskyCT. You are required to submit your homework on HuskyCT as a single pdf file by the end of the day on Thursday.

Quizzes. There will be 4 quizzes, on September 29, October 20, November 10, and December 8. These take 20 minutes and cover problems from class and previous homework.

Exams. There will be 3 exams during class time. There will be no make-ups or extensions.

The tentative dates of these assessments are:

Exam 1: Tuesday, October 13 (Week 7). Based on material covered in Week1-Week4, that was tested in Hwk1-Hwk4.

Exam 2: Tuesday, November 17 (Week 12). Based on material covered in Week5-Week9, that was tested in Hwk5-Hwk8.

Exam 3: Thursday, December 10 (Week 14). Based on material covered in Week10-Week13, that was tested in Hwk9-Hwk10.

Attendance. You are strongly encouraged to attend all in-person lectures. Typed lecture notes will also be available on HuskyCT, but they are not a substitute for class attendance.

Academic Integrity. It is in your best interest to maintain your academic integrity. Any form of academic dishonesty undermines the goals of this course and devalues the learning process. Academic dishonesty is a serious offense at UConn and can result in a zero grade on an assessment and/or failure in the course.

Final grades. Will be computed via the following grading breakdown:

Homework: 20%

Quizzes: 20%

Exam 1: 20%

Exam 2: 20%

Exam 3: 20%

Grading Scale. Math 3150 is a challenging proof course, so test scores are not as high as in most other math classes. Historically, the class average of the total score in Math 3150 is around 75%, and for in-class exams is around 70%.

The *total scores* will be curved at the end of the semester according to the following rule out of 100 maximum points:

Grade	Letter grade
88-100	A
83-87	A-
77-82	B+
70-76	B
65-69	B-
58-64	C+
50-57	C
45-49	C-
40-44	D
0-39	F

TENTATIVE SCHEDULE

Week 1 (Aug 31 - Sept 4). $\mathbb{N}$, $\mathbb{Q}$, $\mathbb{R}$, Induction. Ch. 1.1, 1.2, 1.3

Week 2 (Sept 7 - Sept 11). Completeness Axiom, Archimedean Property, applications. Read Ch.1.4. Hwk1 due on Sept 10.

Week 3 (Sept 14 - Sept 18). Absolute value, Limits of sequences, Examples using the definition. Read Ch. 1.5, 2.7, 2.8. Hwk 2 due on Sept 17.

Week 4 (Sept 21 - Sept 25). Limit theorems for sequences. Examples. Monotone sequences. Read Ch. 2.9, 2.10. Hwk 3 due on Sept 24.

Week 5 (Sept 28 - Oct 2). Subsequences, Cauchy Sequences, Read Ch. 2.10, 2.11. Take Quiz 1 on Sept 29. Hwk 4 due on Oct 1.

Week 6 (Oct 5 - Oct 9). Some elements of topology (not to be tested). Read Ch 2.13. No hwk this week, study for Exam 1.

Week 7 (Oct 12 - Oct 16). Exam 1 is on Oct 13. Continuity. Limits, properties. Read Ch. 3.17, 3.18, 3.20. Hwk 5 due on Oct 15.

Week 8 (Oct 19 - Oct 23). EVT and IVT. Uniform continuity. Read Ch. 3.18, 3.19. Take Quiz 2 on Oct 20. Hwk 6 due on Oct 22.

Week 9 (Oct 26 - Oct 30). Limits of functions, Differentiation, Rolle and MVT. Read 5.28, 5.29. Hwk 7 due on Oct 29.

Week 10 (Nov 2 - Nov 6). Taylor Theorem, Integration. Read Ch. 5.31, 6.32. Hwk 8 due on Nov 5.

Week 11 (Nov 9 - Nov 13). Properties of the integral. Read Ch. 6.33. Take Quiz 3 on November 10. No hwk this week, study for Exam 2.

Week 12 (Nov 16 - Nov 20). Exam 2 is on Nov 17. MVT. Read Ch. 6.33. Hwk 9 due on Nov 19.

Nov 23 - Nov 27. Thanksgiving break

Week 13 (Nov 30 - Dec 4). FTC. Read Ch 6.34. Hwk 10 due on Dec 3.

Week 14 (Dec 7 - Dec 11). Take Quiz 4 on Dec 8. Exam 3 is on Dec 10.